

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Algebra Toolkit

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

ALGEBRA TOOLKIT · CH2 · L1



ENG. ESLAM AHMED

The Map

This lesson collapses Algebra Toolkit into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Name the target

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Write the first line

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Calculate carefully

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Answer in the requested form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Use index laws

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Factorise expression

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 106 website questions, 106 checked solutions, 3 local PDFs read.

01 Name the target

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebra Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3(2x - 5) = \frac{9 - x}{2}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q7

(a)

This is a reasoning step: write one clear sentence, then continue the calculation.

(b)

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: (a) $m^2 - 3m - 40$, (b) $5y(1 + 4y)$, (c) 1, (d) $x = 3$

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

7 (a) Expand and simplify $(m - 8)(m + 5)$ (2) (b) Factorise fully $5y + 20y^2$ (2) (c) Simplify $(p^2 + 3)^0$ (1) (d) Solve $3(2x - 5) = 9$ 2 -x Show clear algebraic working. x = (4)

YOUR SOLUTION

02 Write the first line

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebra Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{x^9}{x^2} = x^{9-2} = x^7$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q1

(a)

$$\frac{x^9}{x^2} = x^{9-2} = x^7$$

(b)

$$\frac{7^8 \times 7^4}{7^3} = 7^{8+4-3} = 7^9$$

Finish: (a) x^7 , (b) 7^9 .

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 02.

QUESTION

1 (a) Simplify $x \times x^9$ (1) (b) Write $7^7 7^8 4^3$ times as a single power of 7 (2)

YOUR SOLUTION

03 Calculate carefully

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebra Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$2^7 \times 4^5 = 2^7 \times (2^2)^5 = 2^{17}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Nov 2023 P2H Q12

(a)

$$2^7 \times 4^5 = 2^7 \times (2^2)^5 = 2^{17}$$

Use index laws

$$4^x = (2^2)^x = 2^{2x}$$

Finish: (a) $x = \frac{17}{2}$, (b) $25p^4y^{16}$

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

12 2 4 4 7 5 x (a) Calculate the value of x x = (2) (b) Simplify fully $125^6 24^2 3^p y^2$

YOUR SOLUTION

04 Answer in the requested form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebra Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{4}{x-2} - \frac{3}{x+1} = \frac{4(x+1) - 3(x-2)}{(x-2)(x+1)}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q12

(a)

$$\frac{4}{x-2} - \frac{3}{x+1} = \frac{4(x+1) - 3(x-2)}{(x-2)(x+1)}$$

(b)

$$2x(x-5)(x-3) = 2x(x^2 - 8x + 15)$$

Finish: (a) $\frac{x+10}{(x-2)(x+1)}$, (b) $2x^3 - 16x^2 + 30x$

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

12 (a) Express $\frac{4}{2} - \frac{3}{1} + x$ as a single fraction. Give your answer in its simplest form. (3) (b) Expand and simplify $2x(x - 5)(x - 3)$ (3)

YOUR SOLUTION

05 Use index laws

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebra Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$10x^2 + 23x + 12 = (5x + 4)(2x + 3)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q21

Factorise expression

$$10x^2 + 23x + 12 = (5x + 4)(2x + 3)$$

Use index laws

$$\frac{10x^2 + 23x + 12}{4x^2 - 9} = \frac{(5x + 4)(2x + 3)}{(2x - 3)(2x + 3)}$$

Finish: (a) $\frac{5x+4}{2x-3}$, (b) $n = 5y - 1$.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

QUESTION

21 (a) Simplify fully $10x^3 - 12x^2 + 4x - 9$ $2x^3 + 2x^2 + 8x + 4$ $2x^3 + 3x^2 + 5x - n$ times $= +$ (b)
Find an expression for n in terms of x . Show clear algebraic working and simplify your expression. (4)

YOUR SOLUTION

06 Factorise expression

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebra Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$10x^2 + 23x + 12 = (5x + 4)(2x + 3)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q21

Factorise expression

$$10x^2 + 23x + 12 = (5x + 4)(2x + 3)$$

Use index laws

$$\frac{10x^2 + 23x + 12}{4x^2 - 9} = \frac{(5x + 4)(2x + 3)}{(2x - 3)(2x + 3)}$$

Finish: (a) $\frac{5x+4}{2x-3}$, (b) $n = 5y - 1$.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 06.

QUESTION

21 (a) Simplify fully $10^3 23 12 4 9 2 2 \times x \times x + + - (3) 2 2 8 4 2 3 2 5 y y y n$ times = + (b) Find an expression for n in terms of y . Show clear algebraic working and simplify your expression. (4)

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Name the target
"work out", "show", "hence", a familiar exam phrase	Write the first line
"work out", "show", "hence", a familiar exam phrase	Calculate carefully
"work out", "show", "hence", a familiar exam phrase	Answer in the requested form
"work out", "show", "hence", a familiar exam phrase	Use index laws
"work out", "show", "hence", a familiar exam phrase	Factorise expression
FINAL BOTTOM LINE	Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Algebraic Roots & Indices

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

ALGEBRAIC ROOTS & INDICES · CH2 · L2



ENG. ESLAM AHMED

The Map

This lesson collapses Algebraic Roots & Indices into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Use index laws with letters

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use negative powers

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Use fractional powers

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Simplify before substituting

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Use index laws

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Simplify surd

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 49 website questions, 49 checked solutions, 3 local PDFs read.

01 Use index laws with letters

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Roots & Indices question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\sqrt{12} = 2\sqrt{3}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q17

Simplify surd

$$\sqrt{12} = 2\sqrt{3}$$

Simplify surd

$$6 + 2\sqrt{12} = 6 + 4\sqrt{3}$$

Finish: (a) shown, (b) $\frac{t^{10}}{9a^8}$

FORMULA BANK

Use index laws before expanding. Fractional index means root; negative index means reciprocal.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

17 (a) Show that $6 \frac{2}{3} \times 12 \frac{12}{13} \times 4 \frac{3}{4} \times 2 + () = + ()$ Show each stage of your working. (3) (b) Simplify fully $27 \frac{12}{15} \times 2 \frac{3}{4} \times 1 - (3)$

YOUR SOLUTION

02 Use negative powers

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Roots & Indices question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{x^9}{x^2} = x^{9-2} = x^7$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q1

(a)

$$\frac{x^9}{x^2} = x^{9-2} = x^7$$

(b)

$$\frac{7^8 \times 7^4}{7^3} = 7^{8+4-3} = 7^9$$

Finish: (a) x^7 , (b) 7^9 .

FORMULA BANK

Use index laws before expanding. Fractional index means root; negative index means reciprocal.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Use fractional powers

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Roots & Indices question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$10x^2 + 23x + 12 = (5x + 4)(2x + 3)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q21

Factorise expression

$$10x^2 + 23x + 12 = (5x + 4)(2x + 3)$$

Use index laws

$$\frac{10x^2 + 23x + 12}{4x^2 - 9} = \frac{(5x + 4)(2x + 3)}{(2x - 3)(2x + 3)}$$

Finish: (a) $\frac{5x+4}{2x-3}$, (b) $n = 5y - 1$.

FORMULA BANK

Use index laws before expanding. Fractional index means root; negative index means reciprocal.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

21 (a) Simplify fully $10^2 3^{12} 4^9 2^2 \times x \times x + + - (3)^2 2^8 4^2 3^2 5^y y y n$ times = + (b)
Find an expression for n in terms of y. Show clear algebraic working and simplify your expression. (4)

YOUR SOLUTION

04 Simplify before substituting

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Roots & Indices question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\sqrt{12} = 2\sqrt{3}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q17

Simplify surd

$$\sqrt{12} = 2\sqrt{3}$$

Simplify surd

$$6 + 2\sqrt{12} = 6 + 4\sqrt{3}$$

Finish: (a) shown, (b) $\frac{t^{10}}{9a^8}$

FORMULA BANK

Use index laws before expanding. Fractional index means root; negative index means reciprocal.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

17 (a) Show that $6 \frac{2}{3} \times 12 \frac{12}{13} \times 4 \frac{3}{4} \times 2 + () = + ()$ Show each stage of your working. (3) (b) Simplify fully $27 \frac{12}{15} \times 2 \frac{3}{4} \times 2 \frac{3}{4} \times 2 \frac{3}{4}$ at - (3)

YOUR SOLUTION

05 Use index laws

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Roots & Indices question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$10x^2 + 23x + 12 = (5x + 4)(2x + 3)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q21

Factorise expression

$$10x^2 + 23x + 12 = (5x + 4)(2x + 3)$$

Use index laws

$$\frac{10x^2 + 23x + 12}{4x^2 - 9} = \frac{(5x + 4)(2x + 3)}{(2x - 3)(2x + 3)}$$

Finish: (a) $\frac{5x+4}{2x-3}$, (b) $n = 5y - 1$.

FORMULA BANK

Use index laws before expanding. Fractional index means root; negative index means reciprocal.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

QUESTION

21 (a) Simplify fully $10x^3 - 12x^2 + 4x - 9$ $2x^3 + 2x^2 + 8x + 4$ $2x^3 + 3x^2 + 5x - n$ times = + (b)
Find an expression for n in terms of x. Show clear algebraic working and simplify your expression. (4)

YOUR SOLUTION

06 Simplify surd

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Roots & Indices question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\sqrt{12} = 2\sqrt{3}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q17

Simplify surd

$$\sqrt{12} = 2\sqrt{3}$$

Simplify surd

$$6 + 2\sqrt{12} = 6 + 4\sqrt{3}$$

Finish: (a) shown, (b) $\frac{t^{10}}{9a^8}$

FORMULA BANK

Use index laws before expanding. Fractional index means root; negative index means reciprocal.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 06.

QUESTION

17 (a) Show that $6 \frac{2}{3} \times 12 \frac{12}{13} \times 4 \frac{3}{2} + () = + ()$ Show each stage of your working. (3) (b) Simplify fully $27 \frac{12}{15} \times 2 \frac{3}{4} \div ()$ (3)

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Use index laws with letters
"work out", "show", "hence", a familiar exam phrase	Use negative powers
"work out", "show", "hence", a familiar exam phrase	Use fractional powers
"work out", "show", "hence", a familiar exam phrase	Simplify before substituting
"work out", "show", "hence", a familiar exam phrase	Use index laws
"work out", "show", "hence", a familiar exam phrase	Simplify surd

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Expanding Brackets

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

EXPANDING BRACKETS · CH2 · L3

ENG. ESLAM AHMED



The Map

This lesson collapses Expanding Brackets into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Name the target

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Write the first line

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Calculate carefully

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Answer in the requested form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Expand brackets

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Factorise expression

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 20 website questions, 20 checked solutions, 3 local PDFs read.

01 Name the target

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Expanding Brackets question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{4}{x-2} - \frac{3}{x+1} = \frac{4(x+1) - 3(x-2)}{(x-2)(x+1)}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q12

(a)

$$\frac{4}{x-2} - \frac{3}{x+1} = \frac{4(x+1) - 3(x-2)}{(x-2)(x+1)}$$

(b)

$$2x(x-5)(x-3) = 2x(x^2 - 8x + 15)$$

Finish: (a) $\frac{x+10}{(x-2)(x+1)}$, (b) $2x^3 - 16x^2 + 30x$

FORMULA BANK

Every term in one bracket multiplies every term in the other bracket.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

12 (a) Express $4\frac{2}{3} - 1\frac{1}{x} + \frac{1}{x-5}$ as a single fraction. Give your answer in its simplest form. (3) (b) Expand and simplify $2x(x-5)(x-3)$ (3)

YOUR SOLUTION

02 Write the first line

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Expanding Brackets question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$(64p^9q^{12})^{2/3} = 16p^6q^8$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Nov 2021 P2H Q12

(a)

$$(64p^9q^{12})^{2/3} = 16p^6q^8$$

(b)

$$\frac{2}{3x} + \frac{4}{5x} - \frac{9}{10x} = \frac{20 + 24 - 27}{30x} = \frac{17}{30x}$$

Finish: (a) $16p^6q^8$, (b) $\frac{17}{30x}$, (c) $8x^3 - 28x^2 - 60x$

FORMULA BANK

Every term in one bracket multiplies every term in the other bracket.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 02.

QUESTION

12 (a) Simplify $\frac{64p^9q^{12}}{2^3 \cdot 2}$ (b) Write as a single fraction $\frac{2^3 \cdot 4^5 \cdot 9^{10}}{x \cdot x \cdot x} \cdot -$ Give your answer in its simplest form. (2) (c) Expand and simplify $4x(x - 5)(2x + 3)$ Show your working clearly. (3)

YOUR SOLUTION

03 Calculate carefully

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Expanding Brackets question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$w = 5(-2)^2 - (-2)^3 = 20 + 8 = 28$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2HR Q1

Expand brackets

$$w = 5(-2)^2 - (-2)^3 = 20 + 8 = 28$$

Finish: 28, $2p(4p - 1)$, $12t^2 - 8t$, $5x^2 + 18x - 8$

FORMULA BANK

Every term in one bracket multiplies every term in the other bracket.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

1 $w = 5y^2 - y^3$ (a) Work out the value of w when $y = -2$ $w = (2)$ (b) Factorise fully $8p^2 - 2p$ (2) (c) Expand $4t(3t - 2)$ (2) (d) Expand and simplify $(5x - 2)(x + 4)$ (2)

YOUR SOLUTION

04 Answer in the requested form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Expanding Brackets question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{4}{x-2} - \frac{3}{x+1} = \frac{4(x+1) - 3(x-2)}{(x-2)(x+1)}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q12

(a)

$$\frac{4}{x-2} - \frac{3}{x+1} = \frac{4(x+1) - 3(x-2)}{(x-2)(x+1)}$$

(b)

$$2x(x-5)(x-3) = 2x(x^2 - 8x + 15)$$

Finish: (a) $\frac{x+10}{(x-2)(x+1)}$, (b) $2x^3 - 16x^2 + 30x$

FORMULA BANK

Every term in one bracket multiplies every term in the other bracket.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

12 (a) Express $\frac{4}{2} - \frac{3}{1} \times x - - +$ as a single fraction. Give your answer in its simplest form. (3) (b) Expand and simplify $2x(x - 5)(x - 3)$ (3)

YOUR SOLUTION

05 Expand brackets

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Expanding Brackets question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$w = 5(-2)^2 - (-2)^3 = 20 + 8 = 28$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2HR Q1

Expand brackets

$$w = 5(-2)^2 - (-2)^3 = 20 + 8 = 28$$

Finish: 28, $2p(4p - 1)$, $12t^2 - 8t$, $5x^2 + 18x - 8$

FORMULA BANK

Every term in one bracket multiplies every term in the other bracket.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

QUESTION

1 $w = 5y^2 - y^3$ (a) Work out the value of w when $y = -2$ $w = (2)$ (b) Factorise fully $8p^2 - 2p$ (2) (c) Expand $4t(3t - 2)$ (2) (d) Expand and simplify $(5x - 2)(x + 4)$ (2)

YOUR SOLUTION

06 Factorise expression

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Expanding Brackets question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4x^2 - 25y^2 = (2x - 5y)(2x + 5y)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2025 P2HR Q13

(a)

$$4x^2 - 25y^2 = (2x - 5y)(2x + 5y)$$

(b)

$$4x(x + 3)(2x - 5)$$

Finish: (a) $(2x - 5y)(2x + 5y)$, (b) $8x^3 + 4x^2 - 60x$

FORMULA BANK

Every term in one bracket multiplies every term in the other bracket.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 06.

QUESTION

13 (a) Factorise $4x^2y - 2xy^2$. (2) (b) Show that $4x^3 + 5x^2 + 3x + 2$ can be written in the form $ax^3 + bx^2 + cx + 2$ where a , b and c are integers to be found. (3)

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Name the target
"work out", "show", "hence", a familiar exam phrase	Write the first line
"work out", "show", "hence", a familiar exam phrase	Calculate carefully
"work out", "show", "hence", a familiar exam phrase	Answer in the requested form
"work out", "show", "hence", a familiar exam phrase	Expand brackets
"work out", "show", "hence", a familiar exam phrase	Factorise expression

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Factorising

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

FACTORISING · CH₂ · L₄

ENG. ESLAM AHMED



The Map

This lesson collapses Factorising into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Take out the common factor

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Factorise a quadratic

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Use difference of squares

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Check by expanding

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Factorise expression

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 (b)(i)

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 23 website questions, 23 checked solutions, 3 local PDFs read.

01 Take out the common factor

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Factorising question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$(2x^3y^5)^4 = 16x^{12}y^{20}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q9

(a)

$$(2x^3y^5)^4 = 16x^{12}y^{20}$$

(b)(i)

$$x^2 + 5x - 36 = (x + 9)(x - 4)$$

Finish: (a) $16x^{12}y^{20}$, (b)(i) $(x + 9)(x - 4)$, (b)(ii) $x = -9$ or $x = 4$

FORMULA BANK

Look for common factor first, then brackets, then difference of squares or quadratic factorising.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

9 (a) Simplify $(2x - 3y + 5)^4$ (2) (b) (i) Factorise $x^2 + 5x - 36$ (2) (ii) Hence, solve $x^2 + 5x - 36 = 0$ (1)

YOUR SOLUTION

02 Factorise a quadratic

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Factorising question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$(2x^3y^5)^4 = 16x^{12}y^{20}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q9

(a)

$$(2x^3y^5)^4 = 16x^{12}y^{20}$$

(b)(i)

$$x^2 + 5x - 36 = (x + 9)(x - 4)$$

Finish: (a) $16x^{12}y^{20}$, (b)(i) $(x + 9)(x - 4)$, (b)(ii) $x = -9$ or $x = 4$

FORMULA BANK

Look for common factor first, then brackets, then difference of squares or quadratic factorising.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Use difference of squares

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Factorising question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$n^2 - 7n + 12 = (n - 3)(n - 4)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1HR Q9

(a)

This is a reasoning step: write one clear sentence, then continue the calculation.

(b)

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: (a) x^9 , (b) $64y^6$, (c) $(n - 3)(n - 4)$

FORMULA BANK

Look for common factor first, then brackets, then difference of squares or quadratic factorising.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

04 Check by expanding

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Factorising question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$9x^2 - 4y^2 = (3x)^2 - (2y)^2 = (3x - 2y)(3x + 2y)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P1H Q11

(a)

$$9x^2 - 4y^2 = (3x)^2 - (2y)^2 = (3x - 2y)(3x + 2y)$$

(b)

$$\frac{7}{8} - \frac{x+3}{4x} = \frac{7x}{8x} - \frac{2(x+3)}{8x}$$

Finish: (a) $(3x - 2y)(3x + 2y)$, (b) $\frac{5x-6}{8x}$

FORMULA BANK

Look for common factor first, then brackets, then difference of squares or quadratic factorising.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

05 Factorise expression

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Factorising question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$(2x^3y^5)^4 = 16x^{12}y^{20}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q9

(a)

$$(2x^3y^5)^4 = 16x^{12}y^{20}$$

(b)(i)

$$x^2 + 5x - 36 = (x + 9)(x - 4)$$

Finish: (a) $16x^{12}y^{20}$, (b)(i) $(x + 9)(x - 4)$, (b)(ii) $x = -9$ or $x = 4$

FORMULA BANK

Look for common factor first, then brackets, then difference of squares or quadratic factorising.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

06 (b)(i)

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Factorising question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$(2x^3y^5)^4 = 16x^{12}y^{20}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q9

(a)

$$(2x^3y^5)^4 = 16x^{12}y^{20}$$

(b)(i)

$$x^2 + 5x - 36 = (x + 9)(x - 4)$$

Finish: (a) $16x^{12}y^{20}$, (b)(i) $(x + 9)(x - 4)$, (b)(ii) $x = -9$ or $x = 4$

FORMULA BANK

Look for common factor first, then brackets, then difference of squares or quadratic factorising.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 06.

QUESTION

9 (a) Simplify $(2x - 3y + 5)^4$ (2) (b) (i) Factorise $x^2 + 5x - 36$ (2) (ii) Hence, solve $x^2 + 5x - 36 = 0$ (1)

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Take out the common factor
"work out", "show", "hence", a familiar exam phrase	Factorise a quadratic
"work out", "show", "hence", a familiar exam phrase	Use difference of squares
"work out", "show", "hence", a familiar exam phrase	Check by expanding
"work out", "show", "hence", a familiar exam phrase	Factorise expression
"work out", "show", "hence", a familiar exam phrase	(b)(i)

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Completing the Square

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

COMPLETING THE SQUARE · CH2 · L5



ENG. ESLAM AHMED

The Map

This lesson collapses Completing the Square into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Halve the coefficient of x

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Add and subtract the square

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Find turning point form

TRIGGER *"work out", "find", a missing value*

04 Solve using completed square

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Evaluate fraction

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 29 website questions, 29 checked solutions, 3 local PDFs read.

01 Halve the coefficient of x

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Completing the Square question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$7 - 12x - 2x^2 = -2x^2 - 12x + 7$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q23

Solve inequality

$$7 - 12x - 2x^2 = -2x^2 - 12x + 7$$

Finish: $25 - 2(x + 3)^2$

FORMULA BANK

$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + c - \left(\frac{b}{2}\right)^2.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

23 Express $7 - 12x - 2x^2$ in the form $a + b(x + c)^2$ where a , b and c are integers.

YOUR SOLUTION

02 Add and subtract the square

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Completing the Square question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$7 - 12x - 2x^2 = -2x^2 - 12x + 7$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q23

Solve inequality

$$7 - 12x - 2x^2 = -2x^2 - 12x + 7$$

Finish: $25 - 2(x + 3)^2$

FORMULA BANK

$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + c - \left(\frac{b}{2}\right)^2.$$

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Find turning point form

TRIGGER *"work out", "find", a missing value*

BECOMES A Completing the Square question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$28 + 24x - 6x^2 = -6x^2 + 24x + 28$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2025 P2H Q25

(a)

$$28 + 24x - 6x^2 = -6x^2 + 24x + 28$$

(b)

$$y = 52 - 6(x - 2)^2$$

Finish: (a) $52 - 6(x - 2)^2$. For the sketch, show a downward parabola with turning point (2, 52) and y -intercept (0, 28).

FORMULA BANK

$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + c - \left(\frac{b}{2}\right)^2.$$

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

25 (a) Write $28x^2 - 6x + x^3$ in the form $a(x - b)^2 + c$ where a , b and c are integers. (3) (b) On the axes below, sketch the graph of $y = 28x^2 - 6x + x^3$. Show clearly the coordinates of the turning point and the coordinates of the point of intersection of the graph with the y -axis. (3)

YOUR SOLUTION

04 Solve using completed square

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Completing the Square question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$7 - 12x - 2x^2 = -2x^2 - 12x + 7$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q23

Solve inequality

$$7 - 12x - 2x^2 = -2x^2 - 12x + 7$$

Finish: $25 - 2(x + 3)^2$

FORMULA BANK

$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + c - \left(\frac{b}{2}\right)^2.$$

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

23 Express $7 - 12x - 2x^2$ in the form $a + b(x + c)^2$ where a , b and c are integers.

YOUR SOLUTION

05 Evaluate fraction

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Completing the Square question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\sqrt[3]{\frac{1}{x}} = x^{-1/3}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1HR Q19

(a)

$$\sqrt[3]{\frac{1}{x}} = x^{-1/3}$$

Evaluate fraction

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: $m = -\frac{1}{3}$, and $3(x+2)^2 + 7$

FORMULA BANK

$$x^2 + bx + c = (x + \frac{b}{2})^2 + c - (\frac{b}{2})^2.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

19 Given that $13 \times x^m$ (a) find the value of m $m = (1)$ Given that a, b and c are integers, (b) express $3x^2 + 12x + 19$ in the form $a(x + b)^2 + c$ (2)

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Halve the coefficient of x
"work out", "show", "hence", a familiar exam phrase	Add and subtract the square
"work out", "find", a missing value	Find turning point form
"work out", "show", "hence", a familiar exam phrase	Solve using completed square
"work out", "show", "hence", a familiar exam phrase	Evaluate fraction
FINAL BOTTOM LINE	Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Algebraic Fractions

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

ALGEBRAIC FRACTIONS · CH2 · L6



ENG. ESLAM AHMED

The Map

This lesson collapses Algebraic Fractions into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Factorise before cancelling

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use common denominators

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Multiply and divide algebraic fractions

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 State excluded values when needed

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Simplify fraction

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 42 website questions, 42 checked solutions, 3 local PDFs read.

01 Factorise before cancelling

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Fractions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$6y^2 - 5y - 4 = (3y - 4)(2y + 1)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: May 2024 P1H Q12

(a)

$$6y^2 - 5y - 4 = (3y - 4)(2y + 1)$$

(b)

$$\frac{2x + 1}{4x} + \frac{7 - 5x}{3x}$$

Finish: (a) $(3y - 4)(2y + 1)$, (b) $\frac{31 - 14x}{12x}$

FORMULA BANK

Factorise first, cancel common factors, then use common denominator if adding.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

12 (a) Factorise fully $6y^2 - 5y - 4$ (2) (b) Express $2\frac{1}{4} \times 7\frac{3}{5} \times x \times x \times x$ as a single fraction in its simplest form. (3)

YOUR SOLUTION

02 Use common denominators

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Fractions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\left(\frac{4}{2x-5} - \frac{3}{2x-3} \right) \div \frac{9x-4x^3}{6x^2-17x+5}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q24

Use trigonometry

$$\left(\frac{4}{2x-5} - \frac{3}{2x-3} \right) \div \frac{9x-4x^3}{6x^2-17x+5}$$

Use trigonometry

$$\frac{4}{2x-5} - \frac{3}{2x-3} = \frac{4(2x-3) - 3(2x-5)}{(2x-5)(2x-3)}$$

Finish: $\frac{1-3x}{x(2x-3)^2}$

FORMULA BANK

Factorise first, cancel common factors, then use common denominator if adding.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 02.

QUESTION

24 Express $\frac{4 \cdot 2 \cdot 5 \cdot 3 \cdot 2 \cdot 3 \cdot 9 \cdot 4 \cdot 6 \cdot 17 \cdot 5 \cdot 3 \cdot 2 \cdot x \cdot x \cdot x \cdot x \cdot x}{- - +}$ divided by $- - +$ as a single fraction in its simplest form.

YOUR SOLUTION

03 Multiply and divide algebraic fractions

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Fractions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\left(\frac{4}{2x-5} - \frac{3}{2x-3} \right) \div \frac{9x-4x^3}{6x^2-17x+5}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q24

Use trigonometry

$$\left(\frac{4}{2x-5} - \frac{3}{2x-3} \right) \div \frac{9x-4x^3}{6x^2-17x+5}$$

Use trigonometry

$$\frac{4}{2x-5} - \frac{3}{2x-3} = \frac{4(2x-3) - 3(2x-5)}{(2x-5)(2x-3)}$$

Finish: $\boxed{\frac{1-3x}{x(2x-3)^2}}$

FORMULA BANK

Factorise first, cancel common factors, then use common denominator if adding.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

24 Express $\frac{4 \cdot 2 \cdot 5 \cdot 3 \cdot 2 \cdot 3 \cdot 9 \cdot 4 \cdot 6 \cdot 17 \cdot 5 \cdot 3 \cdot 2 \cdot x \cdot x \cdot x \cdot x \cdot x}{- - +}$ divided by $- - +$ as a single fraction in its simplest form.

YOUR SOLUTION

04 State excluded values when needed

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Fractions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{25x^2 - 64}{5x^2 - 13x - 6} \times \frac{x^2 - 8x + 15}{5x + 8} - (x - 7)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2H Q21

Use trigonometry

$$\frac{25x^2 - 64}{5x^2 - 13x - 6} \times \frac{x^2 - 8x + 15}{5x + 8} - (x - 7)$$

Factor

$$25x^2 - 64 = (5x - 8)(5x + 8)$$

Finish:

$$\frac{54}{5x + 2}$$

FORMULA BANK

Factorise first, cancel common factors, then use common denominator if adding.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

21 Write $\frac{25}{64} \times \frac{5}{13} \times \frac{6}{8} \times \frac{15}{5} \times \frac{8}{7} \times \frac{2}{2} \times x \times x \times x \times x \times x$ --- times - + + - - () as a single fraction in its simplest form. Show clear algebraic working.

YOUR SOLUTION

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Fractions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\left(\frac{4}{2x-5} - \frac{3}{2x-3} \right) \div \frac{9x-4x^3}{6x^2-17x+5}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q24

Use trigonometry

$$\left(\frac{4}{2x-5} - \frac{3}{2x-3} \right) \div \frac{9x-4x^3}{6x^2-17x+5}$$

Use trigonometry

$$\frac{4}{2x-5} - \frac{3}{2x-3} = \frac{4(2x-3) - 3(2x-5)}{(2x-5)(2x-3)}$$

Finish:

$$\frac{1-3x}{x(2x-3)^2}$$

FORMULA BANK

Factorise first, cancel common factors, then use common denominator if adding.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

QUESTION

24 Express $\frac{4 \cdot 2 \cdot 5 \cdot 3 \cdot 2 \cdot 3 \cdot 9 \cdot 4 \cdot 6 \cdot 17 \cdot 5 \cdot 3 \cdot 2}{x \cdot x \cdot x \cdot x \cdot x}$ divided by $\frac{-}{- +}$ as a single fraction in its simplest form.

YOUR SOLUTION

06 Simplify fraction

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Fractions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$a = \frac{14}{3x-7}, \quad x = \frac{7}{4y-3}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P2H Q19

Simplify fraction

$$a = \frac{14}{3x-7}, \quad x = \frac{7}{4y-3}$$

Simplify fraction

$$3x-7 = \frac{21}{4y-3} - 7$$

Finish: $a = \frac{4y-3}{3-2y}$

FORMULA BANK

Factorise first, cancel common factors, then use common denominator if adding.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 06.

QUESTION

19 a = 14 3 7 x - x = 7 4 3 y - Express a in the form py q ry s + + where p, q, r and s are integers. Give your answer in its simplest form. a =

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Factorise before cancelling
"work out", "show", "hence", a familiar exam phrase	Use common denominators
"work out", "show", "hence", a familiar exam phrase	Multiply and divide algebraic fractions
"work out", "show", "hence", a familiar exam phrase	State excluded values when needed
"work out", "show", "hence", a familiar exam phrase	Use trigonometry
"work out", "show", "hence", a familiar exam phrase	Simplify fraction

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Rearranging Formulas

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

REARRANGING FORMULAS · CH₂ · L7



ENG. ESLAM AHMED

The Map

This lesson collapses Rearranging Formulas into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Make the subject visible

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Clear fractions first

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Undo operations in order

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Factor out the subject

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Rearrange formula

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Square both sides

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 19 website questions, 19 checked solutions, 3 local PDFs read.

01 Make the subject visible

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Rearranging Formulas question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$d = g + 2ac$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q3

(a)

$$d = g + 2ac$$

(b)

$$9ef - 12f = 3f(3e - 4)$$

Finish: $a = \frac{d-g}{2c}, 3f(3e-4), x^2 - 3x - 10, n^6$

FORMULA BANK

Undo operations around the subject, or multiply away fractions first.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

- 3 (a) Make a the subject of $d = g + 2ac$ (2) (b) Factorise fully $9e^2f - 12ef^2$ (2) (c) Expand and simplify $(x + 2)(x - 5)$ (2) (d) Simplify fully $n^4n^7n^5$ (2)

YOUR SOLUTION

02 Clear fractions first

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Rearranging Formulas question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{9a - 7}{5} - \frac{3a - 7}{4} = 4.55$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2H Q14

(a)

$$\frac{9a - 7}{5} - \frac{3a - 7}{4} = 4.55$$

Multiply by 20

$$4(9a - 7) - 5(3a - 7) = 91$$

Finish: (a) $a = 4$, (b) $c = \frac{8-3p^2}{p^2-a}$

FORMULA BANK

Undo operations around the subject, or multiply away fractions first.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 02.

QUESTION

14 (a) Solve $9.75374a - 4.55 = 4.55$ Show clear algebraic working. $a = (3)$ (b) Make c the subject of the formula $p = ac + 83$ (4)

YOUR SOLUTION

03 Undo operations in order

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Rearranging Formulas question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$A = \frac{c}{y} - 5z$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1HR Q6

(a)

$$A = \frac{c}{y} - 5z$$

(b)

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: $c = y(A + 5z)$, 1, $(x - 3)(x - 8)$

FORMULA BANK

Undo operations around the subject, or multiply away fractions first.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

6 (a) Make c the subject of $A = c y - 5z$ (2) (b) Write down the value of g 0 (1) (c) Factorise $x^2 - 11x + 24$ (2)

04 Factor out the subject

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Rearranging Formulas question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$d = g + 2ac$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q3

(a)

$$d = g + 2ac$$

(b)

$$9ef - 12f = 3f(3e - 4)$$

Finish: $a = \frac{d-g}{2c}, 3f(3e-4), x^2 - 3x - 10, n^6$

FORMULA BANK

Undo operations around the subject, or multiply away fractions first.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

- 3 (a) Make a the subject of $d = g + 2ac$ (2) (b) Factorise fully $9e^2f - 12ef^2$ (2) (c) Expand and simplify $(x + 2)(x - 5)$ (2) (d) Simplify fully $n^4n^7n^5$ (2)

YOUR SOLUTION

05 Rearrange formula

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Rearranging Formulas question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = \sqrt{\frac{x+1}{x-4}}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q16

Rearrange formula

$$y = \sqrt{\frac{x+1}{x-4}}$$

Square both sides

$$y^2 = \frac{x+1}{x-4}$$

Finish: $x = \frac{4y^2+1}{y^2-1}$

FORMULA BANK

Undo operations around the subject, or multiply away fractions first.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

06 Square both sides

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Rearranging Formulas question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = \sqrt{\frac{x+1}{x-4}}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q16

Rearrange formula

$$y = \sqrt{\frac{x+1}{x-4}}$$

Square both sides

$$y^2 = \frac{x+1}{x-4}$$

Finish: $x = \frac{4y^2+1}{y^2-1}$

FORMULA BANK

Undo operations around the subject, or multiply away fractions first.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Make the subject visible
"work out", "show", "hence", a familiar exam phrase	Clear fractions first
"work out", "show", "hence", a familiar exam phrase	Undo operations in order
"work out", "show", "hence", a familiar exam phrase	Factor out the subject
"work out", "show", "hence", a familiar exam phrase	Rearrange formula
"work out", "show", "hence", a familiar exam phrase	Square both sides
FINAL BOTTOM LINE	Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Algebraic Proof

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

ALGEBRAIC PROOF · CH2 · L8



ENG. ESLAM AHMED

The Map

This lesson collapses Algebraic Proof into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Start from the expression

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Expand and collect

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Use parity or multiple language

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Finish with the required statement

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Subtract

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 42 website questions, 42 checked solutions, 3 local PDFs read.

01 Start from the expression

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Proof question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x = 0.5727272 \dots$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q13

(a) Let

$$x = 0.5727272 \dots$$

Calculate value

$$1000x = 572.727272 \dots$$

Finish: (a) shown, (b) $\frac{6+3\sqrt{y}}{4-y}$

FORMULA BANK

Start with an expression, expand/simplify until the required form appears.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

13 (a) Use algebra to show that $0.57 \cdot 2 = 63 \cdot 110$ (2) Given that y is a prime number, (b) express $3 \cdot 2 - y$ in the form $a \cdot b \cdot y \cdot c \cdot y + \dots$ where a, b and c are integers. (2)

YOUR SOLUTION

02 Expand and collect

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Proof question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$(n+1)^2 - n^2 = n^2 + 2n + 1 - n^2 = 2n + 1$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q17

Let the two consecutive square

$$(n+1)^2 - n^2 = n^2 + 2n + 1 - n^2 = 2n + 1$$

is one more than an

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: The difference is always odd.

FORMULA BANK

Start with an expression, expand/simplify until the required form appears.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Use parity or multiple language

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Proof question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$N = 5k$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q25

Use trigonometry

$$N = 5k$$

where is an integer

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: $A^2 - B^2$ is always a multiple of 20.

FORMULA BANK

Start with an expression, expand/simplify until the required form appears.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

25 N is a multiple of 5 $A = N + 1$ $B = N - 1$ Prove, using algebra, that $A^2 - B^2$ is always a multiple of 20

YOUR SOLUTION

04 Finish with the required statement

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Proof question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$N = 5k$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q25

Use trigonometry

$$N = 5k$$

where is an integer

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: $A^2 - B^2$ is always a multiple of 20.

FORMULA BANK

Start with an expression, expand/simplify until the required form appears.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

25 N is a multiple of 5 $A = N + 1$ $B = N - 1$ Prove, using algebra, that $A^2 - B^2$ is always a multiple of 20

YOUR SOLUTION

05 Subtract

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Proof question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x = 0.5727272 \dots$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q13

(a) Let

$$x = 0.5727272 \dots$$

Calculate value

$$1000x = 572.727272 \dots$$

Finish: (a) shown, (b) $\frac{6+3\sqrt{y}}{4-y}$

FORMULA BANK

Start with an expression, expand/simplify until the required form appears.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

QUESTION

13 (a) Use algebra to show that $0.57 \cdot 2 = 63 \cdot 110$ (2) Given that y is a prime number, (b) express $3 \cdot 2 - y$ in the form $a \cdot b \cdot y \cdot c \cdot y + \dots$ where a, b and c are integers. (2)

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Start from the expression
"work out", "show", "hence", a familiar exam phrase	Expand and collect
"work out", "show", "hence", a familiar exam phrase	Use parity or multiple language
"work out", "show", "hence", a familiar exam phrase	Finish with the required statement
"work out", "show", "hence", a familiar exam phrase	Subtract

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Solving Linear Equations

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

SOLVING LINEAR EQUATIONS · CH2 · L9



ENG. ESLAM AHMED

The Map

This lesson collapses Solving Linear Equations into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Collect like terms

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Solve with brackets

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Form the equation from words

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Check in the original

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Solve equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Solve quadratic equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 14 website questions, 14 checked solutions, 3 local PDFs read.

01 Collect like terms

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Linear Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3(2x - 5) = \frac{9 - x}{2}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q7

(a)

This is a reasoning step: write one clear sentence, then continue the calculation.

(b)

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: (a) $m^2 - 3m - 40$, (b) $5y(1 + 4y)$, (c) 1, (d) $x = 3$

FORMULA BANK

Expand, collect x -terms on one side, numbers on the other, then divide.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

7 (a) Expand and simplify $(m - 8)(m + 5)$ (2) (b) Factorise fully $5y + 20y^2$ (2) (c) Simplify $(p^2 + 3)^0$ (1) (d) Solve $3(2x - 5) = 9$ $2 - x$ Show clear algebraic working. $x = (4)$

YOUR SOLUTION

02 Solve with brackets

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Linear Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3(2x - 5) = \frac{9 - x}{2}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q7

(a)

This is a reasoning step: write one clear sentence, then continue the calculation.

(b)

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: (a) $m^2 - 3m - 40$, (b) $5y(1 + 4y)$, (c) 1, (d) $x = 3$

FORMULA BANK

Expand, collect x -terms on one side, numbers on the other, then divide.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

7 (a) Expand and simplify $(m - 8)(m + 5)$ (2) (b) Factorise fully $5y + 20y^2$ (2) (c) Simplify $(p^2 + 3)^0$ (1) (d) Solve $3(2x - 5) = 9$ 2 -x Show clear algebraic working. x = (4)

YOUR SOLUTION

03 Form the equation from words

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Linear Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3(2x - 5) = \frac{9 - x}{2}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q7

(a)

This is a reasoning step: write one clear sentence, then continue the calculation.

(b)

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: (a) $m^2 - 3m - 40$, (b) $5y(1 + 4y)$, (c) 1, (d) $x = 3$

FORMULA BANK

Expand, collect x -terms on one side, numbers on the other, then divide.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

7 (a) Expand and simplify $(m - 8)(m + 5)$ (2) (b) Factorise fully $5y + 20y^2$ (2) (c) Simplify $(p^2 + 3)^0$ (1) (d) Solve $3(2x - 5) = 9$ $2 - x$ Show clear algebraic working. $x = (4)$

YOUR SOLUTION

04 Check in the original

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Linear Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$(n - 6)(n + 4) = n^2 - 2n - 24$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P1H Q5

(a)

$$(n - 6)(n + 4) = n^2 - 2n - 24$$

(b)

$$2x - 3 = \frac{3x - 5}{4}$$

Finish: (a) $n^2 - 2n - 24$, (b) $x = \frac{7}{5}$

FORMULA BANK

Expand, collect x -terms on one side, numbers on the other, then divide.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

5 (a) Expand and simplify $(n - 6)(n + 4)$ (2) (b) Solve $2x - 3 = 3$ 5 4 x - Show clear algebraic working. $x = (3)$

YOUR SOLUTION

05 Solve equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Linear Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{5a + 8}{3} - \frac{2a + 5}{4} = 23$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2025 P1HR Q15

Solve equation

$$\frac{5a + 8}{3} - \frac{2a + 5}{4} = 23$$

Multiply by

$$4(5a + 8) - 3(2a + 5) = 276$$

Finish: (a) $a = \frac{37}{2}$, (b) $3y^{-1/2}$

FORMULA BANK

Expand, collect x -terms on one side, numbers on the other, then divide.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

QUESTION

15 (a) Solve $5^8 \cdot 3^2 \cdot 5^4 \cdot 2^3 = a$. Show clear algebraic working. $a = (4)$ (b) Express $y^3 \cdot 1$ in the form $c \cdot y^n$ where c and n are numbers to be found. (2)

YOUR SOLUTION

06 Solve quadratic equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Linear Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{4x+5}{3} - \frac{3-2x}{2} = 13$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P2H Q15

(a)

$$\frac{4x+5}{3} - \frac{3-2x}{2} = 13$$

Multiply by 6

$$2(4x+5) - 3(3-2x) = 78$$

Finish: $x = \frac{11}{2}$, and $-\frac{5}{2} \leq y \leq 6$.

FORMULA BANK

Expand, collect x -terms on one side, numbers on the other, then divide.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 06.

QUESTION

15 (a) Solve $4.5 - 3 + 2x + x - 1 = 13$ Show clear algebraic working. $x = (4)$ (b) Solve the inequality $2y^2 - 7y - 30 < 0$ Show your working clearly. (3)

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Collect like terms
"work out", "show", "hence", a familiar exam phrase	Solve with brackets
"work out", "show", "hence", a familiar exam phrase	Form the equation from words
"work out", "show", "hence", a familiar exam phrase	Check in the original
"work out", "show", "hence", a familiar exam phrase	Solve equation
"work out", "show", "hence", a familiar exam phrase	Solve quadratic equation

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Solving Quadratic Equations

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

SOLVING QUADRATIC EQUATIONS · CH2 · L10

ENG. ESLAM AHMED



The Map

This lesson collapses Solving Quadratic Equations into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Factorise then solve

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use the quadratic formula

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Complete the square

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Use graph roots

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Solve quadratic equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Factorise

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 4 website questions, 4 checked solutions, 3 local PDFs read.

01 Factorise then solve

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Quadratic Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x^2 - 5x - 36 = 0$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q6

Solve quadratic equation

$$x^2 - 5x - 36 = 0$$

Factorise

$$(x - 9)(x + 4) = 0$$

Finish: $x = 9$ or $x = -4$.

FORMULA BANK

Factorise, complete square, or use formula. Each bracket equal to zero gives a root.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

02 Use the quadratic formula

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Quadratic Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x^2 - 5x - 36 = 0$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q6

Solve quadratic equation

$$x^2 - 5x - 36 = 0$$

Factorise

$$(x - 9)(x + 4) = 0$$

Finish: $x = 9$ or $x = -4$.

FORMULA BANK

Factorise, complete square, or use formula. Each bracket equal to zero gives a root.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Complete the square

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Quadratic Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x^2 - 21x + 20 = 0$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2HR Q9

Solve quadratic equation

$$x^2 - 21x + 20 = 0$$

Factorise

$$(x - 20)(x - 1) = 0$$

Finish: $x = 20$ or $x = 1$.

FORMULA BANK

Factorise, complete square, or use formula. Each bracket equal to zero gives a root.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

| Solve $x^2 - 7x + 10 = 0$.

YOUR SOLUTION

04 Use graph roots

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Quadratic Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$5x^2 + 9x - 17 = 0$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: May 2025 4WM1H Q15

Solve quadratic equation

$$5x^2 + 9x - 17 = 0$$

Use quadratic formula

$$x = \frac{-9 \pm \sqrt{9^2 - 4(5)(-17)}}{2(5)}$$

Finish: $x = 1.15$ or $x = -2.95$.

FORMULA BANK

Factorise, complete square, or use formula. Each bracket equal to zero gives a root.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

05 Solve quadratic equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Quadratic Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x^2 - 5x - 36 = 0$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q6

Solve quadratic equation

$$x^2 - 5x - 36 = 0$$

Factorise

$$(x - 9)(x + 4) = 0$$

Finish: $x = 9$ or $x = -4$.

FORMULA BANK

Factorise, complete square, or use formula. Each bracket equal to zero gives a root.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

QUESTION

| Solve $x^2 - 7x + 10 = 0$.

YOUR SOLUTION

06 Factorise

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Quadratic Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x^2 - 5x - 36 = 0$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q6

Solve quadratic equation

$$x^2 - 5x - 36 = 0$$

Factorise

$$(x - 9)(x + 4) = 0$$

Finish: $x = 9$ or $x = -4$.

FORMULA BANK

Factorise, complete square, or use formula. Each bracket equal to zero gives a root.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Factorise then solve
"work out", "show", "hence", a familiar exam phrase	Use the quadratic formula
"work out", "show", "hence", a familiar exam phrase	Complete the square
"work out", "show", "hence", a familiar exam phrase	Use graph roots
"work out", "show", "hence", a familiar exam phrase	Solve quadratic equation
"work out", "show", "hence", a familiar exam phrase	Factorise

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Solving Inequalities

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

SOLVING INEQUALITIES · CH2 · L11



ENG. ESLAM AHMED

The Map

This lesson collapses Solving Inequalities into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Name the target

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Write the first line

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Calculate carefully

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Answer in the requested form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Solve inequality

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Factorise

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 22 website questions, 22 checked solutions, 3 local PDFs read.

01 Name the target

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Inequalities question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{4-3x}{5} - \frac{3x-5}{2} = -3$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q19

(a)

$$\frac{4-3x}{5} - \frac{3x-5}{2} = -3$$

Multiply by 10

$$2(4-3x) - 5(3x-5) = -30$$

Finish: $x = 3$, and $-\frac{8}{5} \leq y \leq 5$.

FORMULA BANK

Solve like an equation, but flip the sign when multiplying/dividing by a negative.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

19 (a) Solve $4^3 5^3 5^2 3^{---} = -x^x$ Show clear algebraic working. $x = (3)$ (b) Solve the inequality $5y^2 - 17y < 40$ (3)

YOUR SOLUTION

02 Write the first line

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Inequalities question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$-2 \leq y < 1$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1HR Q3

Solve inequality

$$-2 \leq y < 1$$

use a closed circle at

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: Closed at -2 , open at 1 ; integers $-3, -2, -1, 0, 1, 2$.

FORMULA BANK

Solve like an equation, but flip the sign when multiplying/dividing by a negative.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 02.

QUESTION

3 (a) On the number line, show the inequality $-2 < y < 1$ (2) (b) Write down all the values of n that satisfy $-3.4 < n < 2$ (2)

YOUR SOLUTION

03 Calculate carefully

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Inequalities question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4x(2x + 5) - 3x(2x - 3) = 8x^2 + 20x - 6x^2 + 9x = 2x^2 + 29x$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2H Q5

(a)

$$4x(2x + 5) - 3x(2x - 3) = 8x^2 + 20x - 6x^2 + 9x = 2x^2 + 29x$$

Expand brackets

$$\frac{y^5 \times y^n}{y^6} = y^{n-1} = y^{13}$$

Finish: (a) $2x^2 + 29x$, (b) $n = 14$, (c) $t < 3$

FORMULA BANK

Solve like an equation, but flip the sign when multiplying/dividing by a negative.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

5 (a) Expand and simplify $4x(2x + 5) - 3x(2x - 3)$ (2) Given that $y \times y \times n \times 5 \times 6$ times $= y^{13}$
 (b) work out the value of n . $n = (2)$ (c) (i) Solve the inequality $7t - 8 < 2t + 7$ (2) (ii) On
 the number line below, represent the solution set of the inequality solved in part (c)(i)
 $-5 -4 -3 -2 -1 0 1 2 3 4 5 t$ (1)

YOUR SOLUTION

04 Answer in the requested form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Inequalities question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$-2 \leq n < 3$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1HR Q1

(a)

$$-2 \leq n < 3$$

Calculate value

$$n = -2, -1, 0, 1, 2$$

Finish: $n = -2, -1, 0, 1, 2$; closed at 1, shaded left.

FORMULA BANK

Solve like an equation, but flip the sign when multiplying/dividing by a negative.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

1 n is an integer. (a) Write down all the values of n such that $-2 < n < 3$ (2) (b) On the number line, represent the inequality $y < 1$ -3 -2 -1 0 1 2 3 y (1)

YOUR SOLUTION

05 Solve inequality

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Inequalities question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$-2 \leq y < 1$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1HR Q3

Solve inequality

$$-2 \leq y < 1$$

use a closed circle at

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: Closed at -2 , open at 1 ; integers $-3, -2, -1, 0, 1, 2$.

FORMULA BANK

Solve like an equation, but flip the sign when multiplying/dividing by a negative.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

QUESTION

3 (a) On the number line, show the inequality $-2 < y < 1$ (2) (b) Write down all the values of n that satisfy $-3.4 < n < 2$ (2)

YOUR SOLUTION

06 Factorise

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Inequalities question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{4-3x}{5} - \frac{3x-5}{2} = -3$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q19

(a)

$$\frac{4-3x}{5} - \frac{3x-5}{2} = -3$$

Multiply by 10

$$2(4-3x) - 5(3x-5) = -30$$

Finish: $x = 3$, and $-\frac{8}{5} \leq y \leq 5$.

FORMULA BANK

Solve like an equation, but flip the sign when multiplying/dividing by a negative.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 06.

QUESTION

19 (a) Solve $4^3 5^3 5^2 3^{---} = -x^x$ Show clear algebraic working. $x = (3)$ (b) Solve the inequality $5y^2 - 17y < 40$ (3)

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Name the target
"work out", "show", "hence", a familiar exam phrase	Write the first line
"work out", "show", "hence", a familiar exam phrase	Calculate carefully
"work out", "show", "hence", a familiar exam phrase	Answer in the requested form
"work out", "show", "hence", a familiar exam phrase	Solve inequality
"work out", "show", "hence", a familiar exam phrase	Factorise

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Simultaneous Equations

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

SIMULTANEOUS EQUATIONS · CH2 · L12



ENG. ESLAM AHMED

The Map

This lesson collapses Simultaneous Equations into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Collect like terms

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Solve with brackets

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Form the equation from words

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Check in the original

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Solve simultaneous equations

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 42 website questions, 42 checked solutions, 3 local PDFs read.

01 Collect like terms

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Simultaneous Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3x + 5y = 6$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q7

Solve simultaneous equations

$$3x + 5y = 6$$

Add the equations

$$10x = -5$$

Finish: $x = -\frac{1}{2}, y = \frac{3}{2}$.

FORMULA BANK

Eliminate one letter or substitute, then solve the remaining equation.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

| Solve $3x + 2y = 17$ and $2x - y = 4$.

YOUR SOLUTION

02 Solve with brackets

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Simultaneous Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3x + 5y = 6$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q7

Solve simultaneous equations

$$3x + 5y = 6$$

Add the equations

$$10x = -5$$

Finish: $x = -\frac{1}{2}, y = \frac{3}{2}$.

FORMULA BANK

Eliminate one letter or substitute, then solve the remaining equation.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Form the equation from words

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Simultaneous Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3x + 5y = 6$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q7

Solve simultaneous equations

$$3x + 5y = 6$$

Add the equations

$$10x = -5$$

Finish: $x = -\frac{1}{2}, y = \frac{3}{2}$.

FORMULA BANK

Eliminate one letter or substitute, then solve the remaining equation.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

| Solve $3x + 2y = 17$ and $2x - y = 4$.

04 Check in the original

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Simultaneous Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$2x + 7y = 17$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2HR Q12

Solve simultaneous equations

$$2x + 7y = 17$$

Multiply the first equation by

$$10x + 35y = 85$$

Finish: $x = -2, y = 3$.

FORMULA BANK

Eliminate one letter or substitute, then solve the remaining equation.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

05 Solve simultaneous equations

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Simultaneous Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3x + 5y = 6$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q7

Solve simultaneous equations

$$3x + 5y = 6$$

Add the equations

$$10x = -5$$

Finish: $x = -\frac{1}{2}, y = \frac{3}{2}$.

FORMULA BANK

Eliminate one letter or substitute, then solve the remaining equation.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

QUESTION

| Solve $3x + 2y = 17$ and $2x - y = 4$.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Collect like terms
"work out", "show", "hence", a familiar exam phrase	Solve with brackets
"work out", "show", "hence", a familiar exam phrase	Form the equation from words
"work out", "show", "hence", a familiar exam phrase	Check in the original
"work out", "show", "hence", a familiar exam phrase	Solve simultaneous equations

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Forming & Solving Equations

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

FORMING & SOLVING EQUATIONS · CH2 · L13

ENG. ESLAM AHMED



The Map

This lesson collapses Forming & Solving Equations into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Collect like terms

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Solve with brackets

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Form the equation from words

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Check in the original

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Solve equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 7 website questions, 7 checked solutions, 3 local PDFs read.

01 Collect like terms

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Forming & Solving Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x + 4x + (x - 7) = 137$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q3

Why it moved The question

This is a reasoning step: write one clear sentence, then continue the calculation.

Let the number of buttons

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: 17 buttons.

FORMULA BANK

Name the unknown, translate the sentence/diagram into an equation, then solve and check.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

3 Three tins, A, B and C, each contain buttons. Tin A contains x buttons. Tin B contains 4 times the number of buttons that tin A contains. Tin C contains 7 fewer buttons than tin A. The total number of buttons in the three tins is 137. Work out the number of buttons in tin C.

YOUR SOLUTION

02 Solve with brackets

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Forming & Solving Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4x + 6 = 30 - x$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: May 2025 4WM1HR Q6

Use trigonometry

$$4x + 6 = 30 - x$$

Finish: $x = 4.8$.

FORMULA BANK

Name the unknown, translate the sentence/diagram into an equation, then solve and check.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

6 6 2 10 p (c) Find the value of p p = (1) 5 1 r = (d) Write down the value of r r = (1) 6
The diagram shows triangle ABC A B C Diagram NOT accurately drawn (4x + 6) cm
(30 - x) cm CA x 4 6 cm CB x 30 cm angle angle CAB CBA = Work out the value of x x
=

YOUR SOLUTION

03 Form the equation from words

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Forming & Solving Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x + 4x + (x - 7) = 137$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q3

Why it moved The question

This is a reasoning step: write one clear sentence, then continue the calculation.

Let the number of buttons

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: 17 buttons.

FORMULA BANK

Name the unknown, translate the sentence/diagram into an equation, then solve and check.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

3 Three tins, A, B and C, each contain buttons. Tin A contains x buttons. Tin B contains 4 times the number of buttons that tin A contains. Tin C contains 7 fewer buttons than tin A. The total number of buttons in the three tins is 137. Work out the number of buttons in tin C.

YOUR SOLUTION

04 Check in the original

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Forming & Solving Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4(5 - x)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Nov 2021 P2H Q2

Area of rectangle A

$$4(5 - x)$$

Area of rectangle B

$$5(2x - 1)$$

Finish: $x = 2.5$.

FORMULA BANK

Name the unknown, translate the sentence/diagram into an equation, then solve and check.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

2 Here are two rectangles, rectangle A and rectangle B. Diagram NOT accurately drawn

4 cm	5 cm
(5 - x) cm	(2x - 1) cm

rectangle B rectangle A

The area of rectangle B is twice the area of rectangle A. Work out the value of x. Show your working clearly.

$$X =$$

YOUR SOLUTION

05 Solve equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Forming & Solving Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x + 4x + (x - 7) = 137$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q3

Why it moved The question

This is a reasoning step: write one clear sentence, then continue the calculation.

Let the number of buttons

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: 17 buttons.

FORMULA BANK

Name the unknown, translate the sentence/diagram into an equation, then solve and check.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

3 Three tins, A, B and C, each contain buttons. Tin A contains x buttons. Tin B contains 4 times the number of buttons that tin A contains. Tin C contains 7 fewer buttons than tin A. The total number of buttons in the three tins is 137. Work out the number of buttons in tin C.

YOUR SOLUTION

06 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Forming & Solving Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4x + 6 = 30 - x$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: May 2025 4WM1HR Q6

Use trigonometry

$$4x + 6 = 30 - x$$

Finish: $x = 4.8$.

FORMULA BANK

Name the unknown, translate the sentence/diagram into an equation, then solve and check.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

6 6 2 10 p (c) Find the value of p p = (1) 5 1 r = (d) Write down the value of r r = (1) 6
The diagram shows triangle ABC A B C Diagram NOT accurately drawn (4x + 6) cm
(30 - x) cm CA x 4 6 cm CB x 30 cm angle angle CAB CBA = Work out the value of x x
=

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Collect like terms
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"work out", "show", "hence", a familiar exam phrase	Check in the original
"work out", "show", "hence", a familiar exam phrase	Solve equation
"work out", "show", "hence", a familiar exam phrase	Use trigonometry

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.